

A fifth postulate for the Principia: joint determinacy of place and motion

Result, 2026-09-27; revised the same day after an adversarial Fable review. The user’s goal: identify the statement of the *Principia* whose change gives mechanics with $\hbar > 0$, as the parallel postulate does for curved geometry. The proposed statement is **joint determinacy**: every body admits states in which its place and its quantity of motion are given together, sharply, at one instant. Newton asserts it in the scholium closing Book I, Section I, as his reply to the objection that a vanishing ratio is not ultimate before the quantities vanish and is nothing once they have: the ultimate velocity is “the very velocity with which the body reaches its last place”, a limit “certain and definite” (§1). A place belongs to an instant and a motion to an interval; the classical debate from Zeno’s arrow onward keeps them apart, and Newton’s ultimate velocity joins them.

Joint determinacy can be denied in two ways: in the algebra of observables (they cease to commute) or in the states (a restriction on which states are admissible). For the algebraic denial the analogy with Euclid’s fifth postulate holds in the following precise, partial sense.

1. **Independence (Theorem A).** For every real $\hbar$, the Moyal product $*_{\hbar}$ satisfies Newton’s second law as an identity of observables (Heisenberg form), and for Newton’s integrable cases (inertia, uniform gravity, the force proportional to distance of Book I, Proposition X) every observable evolves by the classical flow. Relative to these Heisenberg-form axioms, commutativity is undecided.
2. **One constant (Theorem B, Gutt’s theorem in this setting).** Every associative product on the polynomial observables that deforms the commutative one, respects complex conjugation, and is covariant under the affine symplectic group (premise (H2), stated and justified in §4) is a Moyal product with one real constant $\hbar$, of the units of action.
3. **The floor (Theorem C).** For $\hbar \neq 0$, every state obeys $\Delta q \Delta p \geq |\hbar|/2$, and every recorded comparison of the inertial line with the constant-force parabola obeys the floor of the Planck paper, Theorem 6, with $|\hbar|$ in place of $\hbar$. For $\hbar = 0$ with all states admissible, point states give zero undetermined disturbance, so there is no floor.
4. **Similarity (Theorem D).** The action-rescaling map $(q, p) \mapsto (\lambda q, \lambda p)$ preserves the product $*_{\hbar}$ only for $\hbar = 0$; up to isomorphism there are two products, commutative and Moyal. This is the mechanical counterpart of Wallis’s form of the fifth postulate (similar figures of every size) and of the absolute length of Lambert and Gauss, in the sense made precise in §6.

What the analogy does not give. The state route shows that noncommutativity is not the only way to a floor: GPT-6 Astra’s routes note proves the same floor in a commutative theory with a Gaussian covariance restriction, with $\hbar_* = 2\zeta$. So “floor if and only if $\hbar \neq 0$ ” holds only with the state-completeness premise of Theorem C(c). Noncommutativity is the unique *algebraic* denial of joint determinacy under (H1)–(H3); it is one of two denials. And (H2) is a premise about the observables, Weyl or Groenewold covariance, which the *Principia* does not supply.

The mathematics is established. Moyal’s product (Moyal 1949, metadata); Groenewold’s reduction of its bracket to the Poisson bracket for quadratic functions (Groenewold 1946, *Physica* 12, 405, metadata) and van Hove’s no-go for quantizing all observables (van Hove 1951, metadata); Vey’s deformations of the Poisson bracket (Vey 1975, metadata); the deformation-quantization programme (Bayen, Flato, Fronsdal, Lichnerowicz and Sternheimer 1978, I, metadata), whose authors framed quantization as a deformation in the spirit of the passage to non-Euclidean and relativistic physics (Flato 1982; Sternheimer 1998; metadata). The uniqueness in Theorem B is Gutt’s theorem: the Moyal product is the unique $Sp(2n, \mathbb{R}) \ltimes \mathbb{R}^{2n}$ -invariant and covariant star product (S. Gutt, *Mém. Acad. Roy. Belg. Cl. Sci.* 44:6, 1983, as cited in

Duval, El Gradechi and Ovsienko 2004, passage); the associativity step is Fletcher’s (Fletcher 1990, metadata); equivalence classes of invariant products are parametrized by Bertelson, Bieliavsky and Gutt (1998) (metadata). Robertson’s inequality is Robertson 1929 (metadata). The contribution here is the identification of Newton’s *velocitas ultima* as the statement concerned, and the reading of Theorems A–D against the parallel postulate, with the limits just stated.

1. The textual anchor

A place is predicated of an instant; a quantity of motion, of an interval. The distinction is the substance of Zeno’s arrow and of Aristotle’s thesis that nothing moves in the now (*Physics* VI; cited by book only, the Greek is not held in this repository). Newton addresses it in the second paragraph of the scholium closing Book I, Section I (1687 text, held in the Latin companion, added 2026-09-27 at the user’s prompting):

Objectio est, quod quantitatum evanescentium nulla sit ultima proportio; quippe quæ, antequam evanuerunt, non est ultima, ubi evanuerunt, nulla est. Sed & eodem argumento æque contendere posset nullam esse corporis ad certum locum pergentis velocitatem ultimam. [...] Per velocitatem ultimam intelligi eam, qua corpus movetur neq; antequam attingit locum ultimum & motus cessat, neq; postea, sed tunc cum attingit, id est illam ipsam velocitatem quacum corpus attingit locum ultimum & quacum motus cessat. [...] Extat limes quem velocitas in fine motus attingere potest, non autem transgredi. Hæc est velocitas ultima. [...] Cumq; hic limes sit certus & definitus, Problema est vere Geometricum eundem determinare.

“The objection is that vanishing quantities have no ultimate proportion, since before they vanish it is not ultimate and when they have vanished it is none. By the same argument one could claim that a body reaching a given place has no ultimate velocity [...]. By the ultimate velocity is meant that with which the body moves neither before it reaches its last place and the motion ceases, nor after, but when it reaches it, that is, the very velocity with which the body reaches its last place [...]. There is a limit which the velocity at the end of the motion can reach and not pass. This is the ultimate velocity. [...] And since this limit is certain and definite, determining it is a truly geometrical problem.”

A velocity, a ratio over an interval, is here assigned as a certain and definite quantity to the instant of arrival at a place. The third paragraph adds that the compared quantities are “semper diminuendas sine limite”, always to be diminished without limit. The scholium defends the lemmas of Section I; Newton applies those lemmas to the sagitta of Lemma X and the polygon of Proposition I, and the Scholium to the Definitions defines motion as “translatio corporis de loco [...] in locum” (location only; the Latin of that scholium is not held here). For a geometrical ratio the thesis is a theorem about limits and holds in every model below. For a *recorded* comparison, where the place is read at an instant and the motion inferred over an interval, it becomes the claim that states exist in which both are sharp. That claim is joint determinacy. What remains interpretive is the extension from the geometrical thesis to records, which Newton did not distinguish.

Earlier authors on the arrow and uncertainty. Bohm’s *Quantum Theory* (Prentice-Hall, 1951, ch. 8, pp. 145–148) discusses Zeno’s arrow while contrasting classical definite position and velocity with the quantum description (passage as reported from a transcription by the GPT-6 Astra bibliography search of 2026-09-27; not checked here). Landsberg, *Seeking Ultimeates* (IOP, 2nd ed., 2000, p. 124) connects velocity over an interval, uncertain position and the arrow (passage as reported). Vaidman (2008) links the arrow’s momentum and localization to Heisenberg’s relation before turning to the quantum Zeno effect (passage as reported; DOI verified). Goyal (2026), §VI.1, ties the arrow to Bohr’s complementarity of coordination and causality (passage as reported; DOI verified). Wagstaff (*Philosophy Now* 45, 2004) argues

that exact position and exact momentum cannot be represented at once, since one needs an instant and the other an interval (passage). Lynds (2003) proposes an indeterminacy that enables motion and distinguishes it explicitly from the indeterminacy of h (passage as reported). The philosophical debate on instantaneous velocity is Arntzenius (2000) and Smith (2003) (abstracts). The quantum Zeno effect of Misra and Sudarshan is a different phenomenon. No author found connects Newton's *velocitas ultima* to uncertainty; that is a limit of the search, not a claim of absence.

2. The family of products

Observables of one degree of freedom (the transverse coordinate of Galileo's comparison) are polynomials in q and p with complex coefficients. Write $\mu(f \otimes g) = fg$ and

$$P = \partial_q \otimes \partial_p - \partial_p \otimes \partial_q, \quad f *_h g = \mu \circ \exp\left(\frac{i\hbar}{2}P\right)(f \otimes g),$$

a finite sum on polynomials. For $\hbar = 0$ it is the commutative product; $q *_h p - p *_h q = i\hbar$. The Moyal bracket is $\{f, g\}_h = (f *_h g - g *_h f)/(i\hbar) = \frac{2}{\hbar} \mu \circ \sin(\frac{\hbar}{2}P)(f \otimes g)$, with $\{f, g\}_0$ the Poisson bracket.

3. Theorem A: independence from the Laws in Heisenberg form

Theorem A. Let $H = p^2/(2M) + V(q)$ with V a polynomial and $M > 0$. For every real $\hbar$:

- (a) $\{q, H\}_h = p/M$ and $\{p, H\}_h = -V'(q)$: the Heisenberg equations are Newton's second law, $M\ddot{q} = -V'(q)$, as an identity of observables;
- (b) if $\deg V \leq 2$ (inertia $V = 0$, uniform gravity $V = -Fq$, the force proportional to distance of Book I, Proposition X, $V = kq^2/2$), then $\{f, H\}_h = \{f, H\}_0$ for every f : every observable evolves by the classical flow φ_t , and each φ_t is an automorphism of $*_h$.

Proof. The bracket $\{f, g\}_h$ is the Poisson bracket plus terms $\hbar^{2j} \mu P^{2j+1}(f \otimes g)$, $j \geq 1$, each of which puts at least three derivatives on each factor; they vanish if one factor has degree at most two. This gives (a) because q and p have degree one, and (b) because H has degree two. The flow of a quadratic Hamiltonian is an affine symplectic map, under which $*_h$ is covariant: translations commute with the constant-coefficient operator P , and a linear map of determinant one leaves P invariant. $\square$

The independence is relative to axioms stated in Heisenberg form: the Laws as identities of observables and the classical evolution of Newton's integrable cases. Both the commutative product and every Moyal product satisfy them.

4. Theorem B: one action constant, under covariance

Consider products $f * g = \mu \circ B(f \otimes g)$ on $\mathbb{C}[q, p]$ with B a bilinear map. On polynomials every bilinear map can be written as a formal bidifferential operator $B = \sum c_{\alpha\beta}(q, p) \partial^\alpha \otimes \partial^\beta$, so this is no restriction. Assume:

- (H1) $*$ is associative with unit 1, and $c_{00} = 1$;
- (H2) $*$ is covariant under the affine symplectic group: for every translation and every linear map A of determinant one, $(f \circ A) * (g \circ A) = (f * g) \circ A$;
- (H3) $\bar{f} * \bar{g} = \bar{g} * \bar{f}$.

What (H2) says, and what it does not follow from. By Theorem A(b), the flows of Newton's integrable cases are affine symplectic maps: free flight is a shear, uniform gravity adds translations, and the harmonic flow is a rotation in suitable units; shears and rotations generate

the group of linear maps of determinant one. (H2) requires that these flows act by automorphisms of the product, that is, that quadratic observables evolve classically in the deformed theory as well. The Laws alone do not impose it. The product $f * g = \mu \circ \exp(\frac{i\hbar}{2}P + \sigma \partial_q \otimes \partial_q)(f \otimes g)$, $\sigma \in \mathbb{R}$, is associative and unital, satisfies (H3), is covariant under translations and shears, and satisfies Theorem A(a) verbatim; it fails only under the harmonic rotation, where the product qp drifts by a term proportional to σk (reviewer's counterexample). So (H2) is a premise of Weyl or Groenewold covariance, physically natural (the kinematics of composing observables is the same along Newton's integrable motions) and outside the *Principia*.

Theorem B. Under (H1)–(H3), $*$ is $*_{\hbar}$ for a unique real $\hbar$, of the units of action.

Proof. Translation covariance makes the coefficients $c_{\alpha\beta}$ constant. Then B is a formal power series in the components of two vectors $u = \partial^{(1)}$, $v = \partial^{(2)}$, invariant under the simultaneous action of the determinant-one linear maps; by the first fundamental theorem of invariant theory for $SL(2)$, it is a function of $\det(u, v) = P$ alone: $B = F(P)$, $F(0) = 1$. For three factors put $x = P_{12}$, $y = P_{13}$, $z = P_{23}$; the three 2×2 minors of a 2×3 matrix are algebraically independent, and associativity reads $F(x)F(y+z) = F(z)F(x+y)$ as formal series. At $z = 0$ this is $F(x)F(y) = F(x+y)$, so $F(x) = e^{\kappa x}$ (Fletcher's step). Exchanging the factors reverses P and conjugation leaves it real, so (H3) gives $\bar{\kappa} = -\kappa$: $\kappa = i\hbar/2$ with $\hbar$ real. P carries the reciprocal units of qp . $\square$

Two products up to isomorphism. The dilation $D_{\lambda}(q, p) = (\lambda q, \lambda p)$ satisfies $P((f \circ D_{\lambda}) \otimes (g \circ D_{\lambda})) = \lambda^2(P(f \otimes g)) \circ D_{\lambda}$, so $f \mapsto f \circ D_{\lambda}$ is an isomorphism from $*_{\lambda^2\hbar}$ to $*_{\hbar}$, and $(q, p) \mapsto (q, -p)$ carries $*_{\hbar}$ to $*_{-\hbar}$. All products with $\hbar \neq 0$ are isomorphic, and distinct from the commutative one. Geometry has three constant-curvature classes; this algebraic family has two.

5. Theorem C: the floor on the recorded comparison

A state is a linear functional ω with $\omega(1) = 1$ and $\omega(\bar{f} *_{\hbar} f) \geq 0$; it is regular if its GNS representation integrates to Weyl operators.

Theorem C. (a) For $\hbar \neq 0$ every state satisfies $\Delta q \Delta p \geq |\hbar|/2$.

- (b) For $\hbar \neq 0$, consider the comparison of the inertial line with the constant-force parabola over a cell of duration τ , with fall $s = F\tau^2/(2M)$ and impulse $J = F\tau$ (the Planck paper's convention), made by any instrument coupled unitarily to the body, with regular joint states, deciding between the two hypotheses for every initial state with error probability at most $\epsilon < \frac{1}{2}$. Then the undetermined impulses $\hat{D}_j$ and displacements $\hat{X}_j$, with spreads taken as suprema over the interpolating states of the Planck paper, satisfy

$$\frac{s}{8} \sum_j \Delta(\hat{D}_j) + \frac{J}{2} \sum_j \Delta(\hat{X}_j) \geq |\hbar| \arcsin(1 - 2\epsilon) > 0.$$

- (c) For $\hbar = 0$, if every probability measure on phase space is an admissible state of body and pointers (state completeness), there is no floor: a pointer prepared at $p_y = 0$ and coupled by gqp_y reads q with delivered impulse $D = 0$ and displacement $X = 0$.

Proof. (a) Cauchy–Schwarz for the positive form $(f, g) \mapsto \omega(\bar{f} *_{\hbar} g)$ applied to $q - \omega(q)$ and $p - \omega(p)$, with $q *_{\hbar} p - p *_{\hbar} q = i\hbar$ (Robertson). (b) By the Stone–von Neumann theorem a regular representation is a multiple of the Schrödinger representation with constant $\hbar$; the multiplicity space is absorbed into the apparatus. Theorem 6 of the Planck paper is proved from the Weyl relations and Mandelstam–Tamm in the Bures angle, which is independent of the representation, under exactly these quantifiers; for $\hbar < 0$ apply the reflection $(q, p) \mapsto (q, -p)$. (c) The coupling moves the pointer by gq and kicks the body by $-gp_y = 0$. $\square$

The floor bounds the part of the comparison that Newton's scholium says is to be diminished without limit: the recorded fall and impulse. Part (c) shows that the floor needs either noncommutativity or a restriction of states; Astra's routes note realizes the second in a commutative theory with the same constants.

6. Theorem D: similarity

Theorem D. For $\lambda \neq \pm 1$, D_λ is an automorphism of the associative product $*_{\hbar}$ if and only if $\hbar = 0$.

Proof. D_λ carries $*_{\lambda^2\hbar}$ to $*_{\hbar}$, and $*_{\lambda^2\hbar} = *_{\hbar}$ iff $\lambda^2\hbar = \hbar$. $\square$

The statement concerns the product, the structure that records see. D_λ also multiplies the Poisson bracket by λ^2 , so it is not an automorphism of the classical Poisson algebra either; for the dynamics it is a similarity that rescales actions, in both theories. The honest form of the analogy is therefore the one stated in the dimensional note, Theorem B: similar records at every scale exist exactly when no absolute action is available. Wallis replaced Euclid's fifth postulate by the existence of similar figures of different size; Lambert and Gauss saw that its negation brings an absolute length. Theorem D is the counterpart for the product of observables, with $\hbar$ as the absolute action.

7. What remains, and consequence for STATE

Theorems A–D are proved for one degree of freedom; several degrees follow with $Sp(2n)$, whose invariants of two vectors are again generated by the symplectic pairing. The analogy with the parallel postulate holds for the algebraic route under (H1)–(H3): independence from the Heisenberg-form axioms, a unique deformation with one constant, and a floor. It is partial in two ways stated above: (H2) is an added premise, and the state route gives the same floor without noncommutativity. The identification with Newton rests on his *velocitas ultima*, with the step from geometry to records interpretive.

For STATE item 2 this places the necessity question precisely. Theorem U supplies the value of the unit from radiation thermodynamics; Theorem I and the routes note say what must be denied; this note shows that the algebraic denial is a single step with a single constant, and that Newton asserted, in the ultimate velocity, the joint determinacy it denies.